

How to connect to Linux Machine remotely?

To connect to a remote Linux machine, we must have a tool on your laptop/desktop called -**SSH Client**

Windows users:

- gitbash
- putty
- mobaXterm
- Window PowerShell

What we need to connect to the remote server?

- IP address
- Create a key-pair (user name password)
- Allow Security Group Access

ssh -i key.pem ubuntu@<public ip>



What is a Server?

Where you host an application, you send a request and get the response.

Computer which allows to run our application.

Use to run software/application – request and response

What is an Application/S/W?

Piece of code serves some business requirements

Basics...



Server



Applications

Basics... Application Types

- 1. Standalone Applications:**
applications we install on **Desktops** – only a **single user** can access the application at any point of time
ex: VLC player, calc app, MS Office etc..
- 2. Web Applications:**
applications that can be accessed by N number of people from N number of Geographic Locations over an internet.
ex: simple websites like ThinkWright.in, simplesite.com etc..
These application we host on **web servers** such as **apache, nginx, iis, tomcat etc...**
- 3. Enterprise Applications:**
applications that can be accessed by N number of people from N number of Geographic Locations over a internet, but have an extended features over web applications such as **security, messaging, High Availability, Integrations etc..**
ex: amazon.com, bookmyshow, flipkart, etc...
These type of application we host / install on **application servers** such as **WebSphere, Jboss, Weblogic etc...**

Type of Applications:

Stand-alone applications:

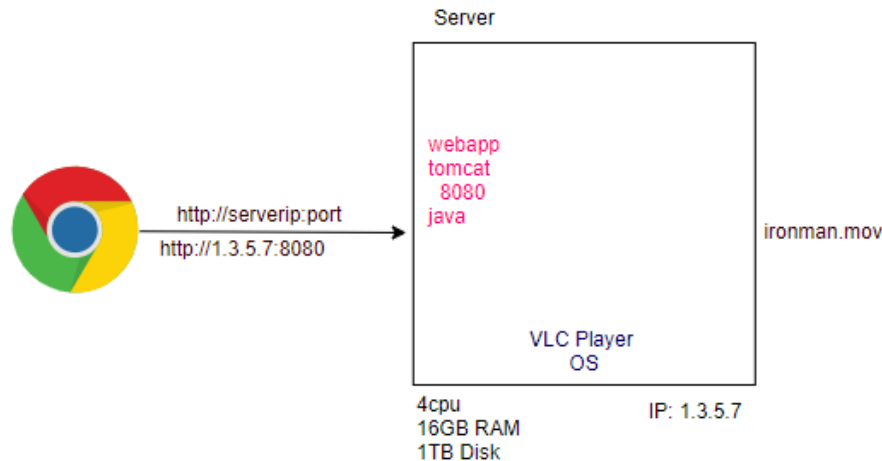
ex: paint/ ms office / calc / vlc player

Web application: Any application that can be opened in a browser

ex: amazon.in /facebook.com

web server needs a runtime to run them

web servers: tomcat /Apache/ IIS/ nginx



What is the difference between webserver and app server?

App server can also provide the business logic, messaging, security features, an extension of Web Server.

Software Engineering?

Our main goal is developing an S/W, deliver the S/W to the customer and maintain the S/W, ensure that must be used by end users.

Software Engineering

"The systematic, disciplined approach to the development, operation and maintenance of software"

-- IEEE

Goal:

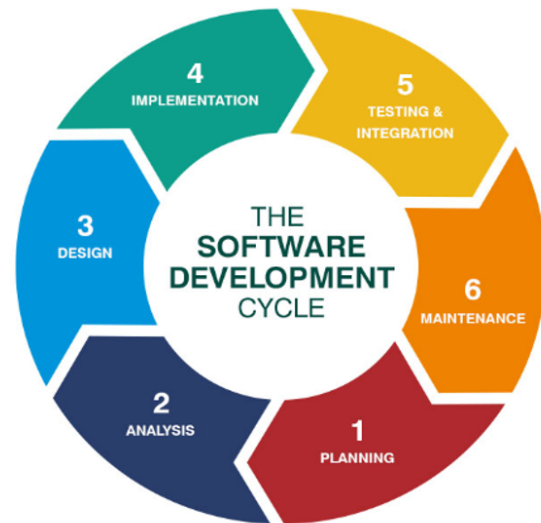
Is to deliver an efficient and reliable software product to client OR to end user OR to market.

To deliver the same, we are following some processes called SDLC models.

SDLC

SDLC defines the complete life cycle of software i.e. all the tasks involved in planning, creating, testing, and delivering a Software to client or market.

SDLC process aims to produce high-quality software that meets customer expectations



We have different SDLC models... Various SDLC models in IT

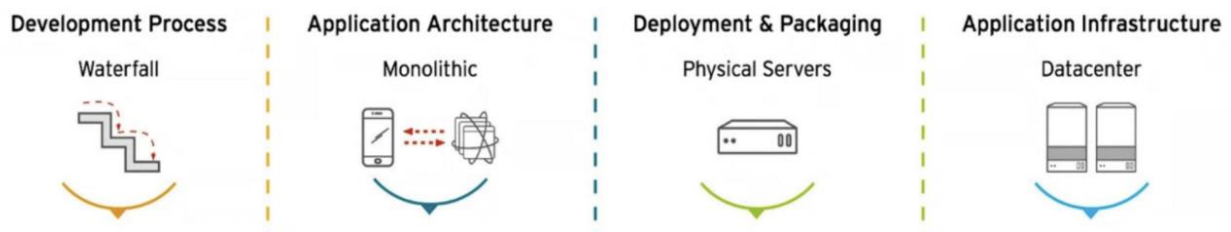
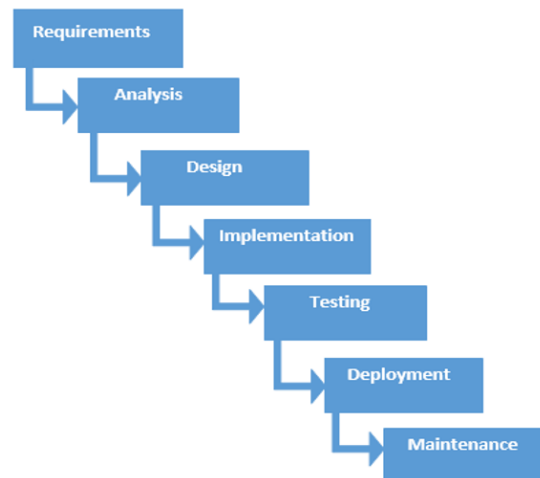
Evolution of IT

Waterfall

New phase in the development process begins only if the previous phase is complete

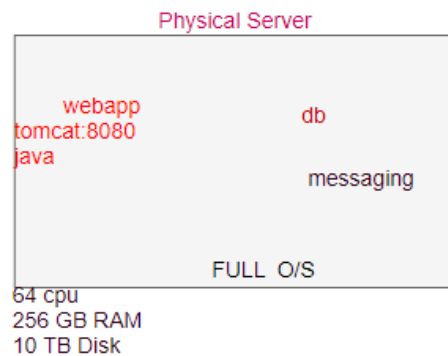
Challenges:

- ✓ Time to market is very high
- ✓ No client interaction
- ✓ Requirements are rigid, means we cannot accommodate dynamic requirements.



Monolithic Architecture (Tightly Coupled Architecture)

Many components bundled and deployed together (single artifact).



- Monolithic Architecture (Tightly Coupled Architecture)
- Heavy Weight
- Resource Wastage
- Maintenance Cost
- Scalability
- Troubleshooting
- Delivery time
- Size of application big in size.
- Its bigger, it can not be shipped easily.
- installation time/ deployment time takes more time.
- troubleshooting is big problem.

All the dependent components tightly coupled and bundled together, since they are bundled together, they are heavy, it is heavy weight it takes longer time to install, difficult to deploy together must be executed together to run the application smoothly.

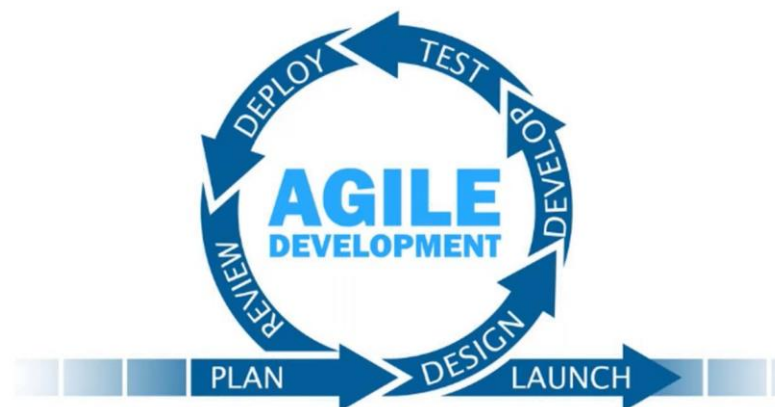
Data Centers:

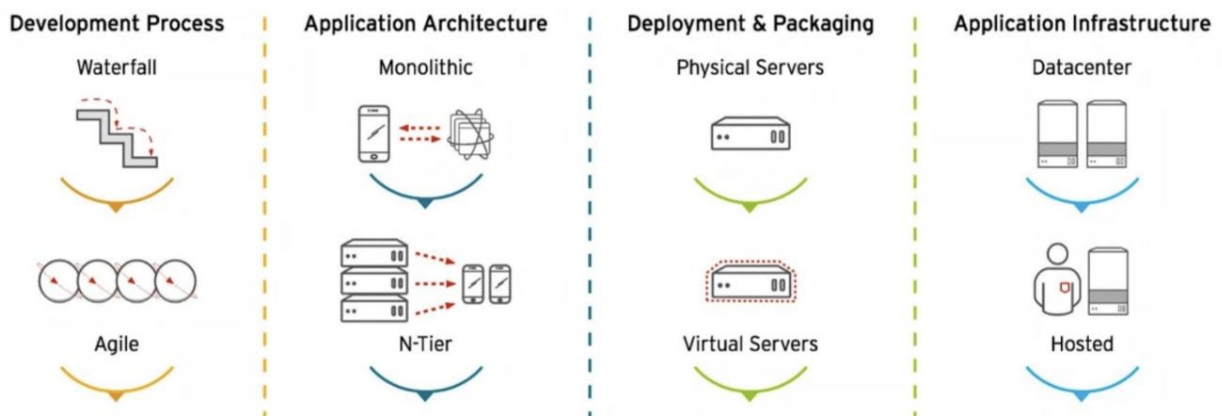
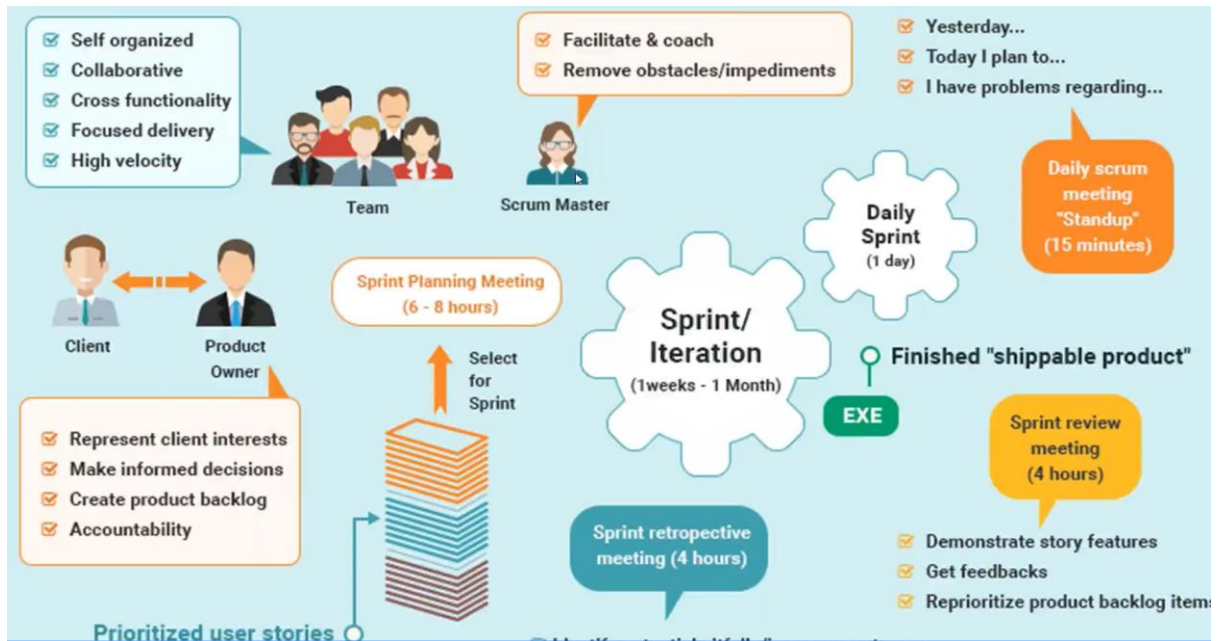
Organization's building where they kept their all H/W servers, like a home kitchen/self-home.

Current Market demand is speed...

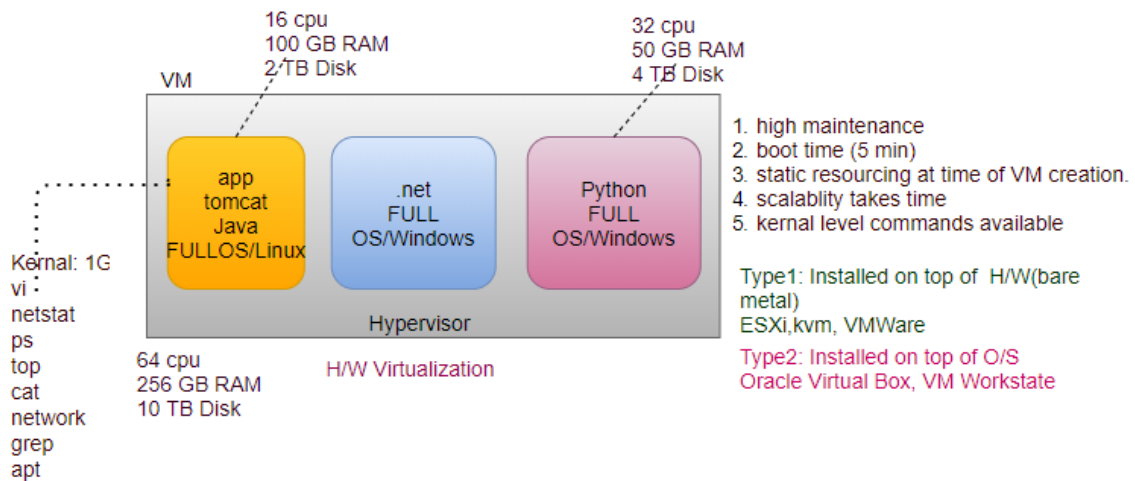
Agile :(Iterative and Incremental)

Evolution of IT





Virtual Servers:



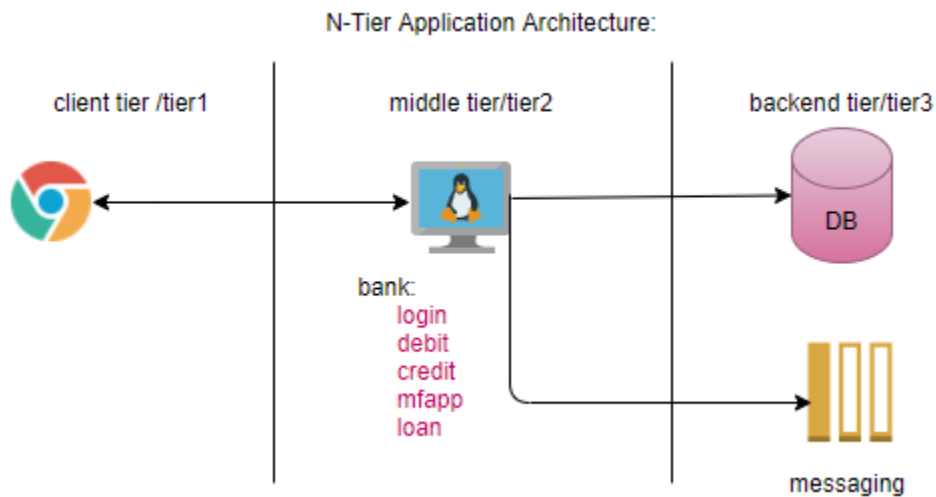
Virtual Server: Hypervisor, Convert Big Fat server into multiple small virtual servers, each virtual machine has their own O/S, same time we have a resource shared among these virtual machines.

Type1: install hypervisor on top of bare metal. During creation of virtual machine asks for allocated resources, e.g... **ESXi, VMware, kvm**

Type2: Hypervisor installed on the top of O/S (para virtualization), e.g., **Oracle Virtual Box**

N-Tier Application Architecture:

MVC (decoupled architecture): presentation layer in different server, business layer in different server, backend layer in different server, e.g., Oracle



Hosted:

Third party data centers. Like a rented house /hotel either eat or not have to pay.

Benefits:

- Development speed increased.
- Utilization of resources more effectively.
- Light weight components easy to deploy and manage.

Challenges at Operation side...

Infrastructure Readiness:

Infrastructure is not ready compare to the development, e.g., we deploy higher version features on lower version of server.

Scalability:

Big Friday Salary (10-100 servers) how quickly can scale my environment. Lots of effort required. Process takes longer time to create VM.

Quality of the Product:

Due to tight delivery schedule, compromise with quality, e.g. sprint is ending tomorrow and development is completed today.

Limited Release Window- Release may slip/fail:

Client don't allow to deploy during peak business hours, only sat-sun allowed. How to deliver without any downtime.

What is the Need?

- Shorter Deployment Cycles, Faster Innovation.
- Reduced Deployment Failures, Rollbacks, and Time to Recover
- Improved Communication and Collaboration
- Increased Efficiency.

Approach?

- Infrastructure readiness
- Scalability
- Quality of the product (Automation)
- Limited Release Windows
 - We don't want downtime (CICD)

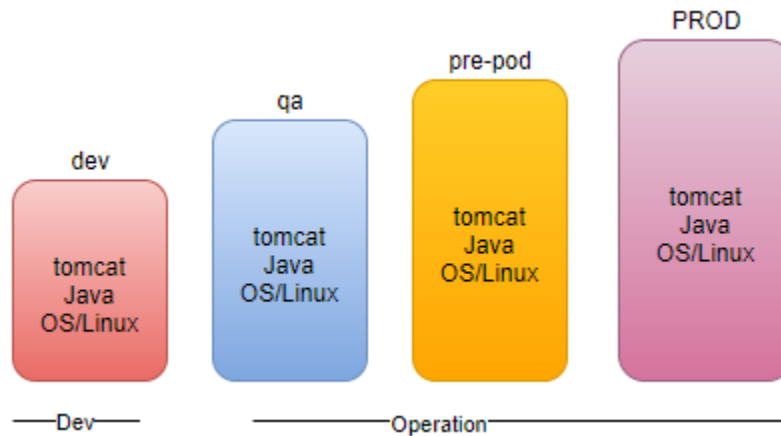
Problem Statement

Application is running fine in development environment but not working in QA environment.

Reason:

- Different versions
- Dependencies

There is pre-requisite of environment must be ready (O/S, Java, tomcat) on all the environments.



We are delivering **only artifacts (.war)** but **not the dependencies**. It may be possible that in different environment different version of java running, causes application is running in one environment but not in other environment.

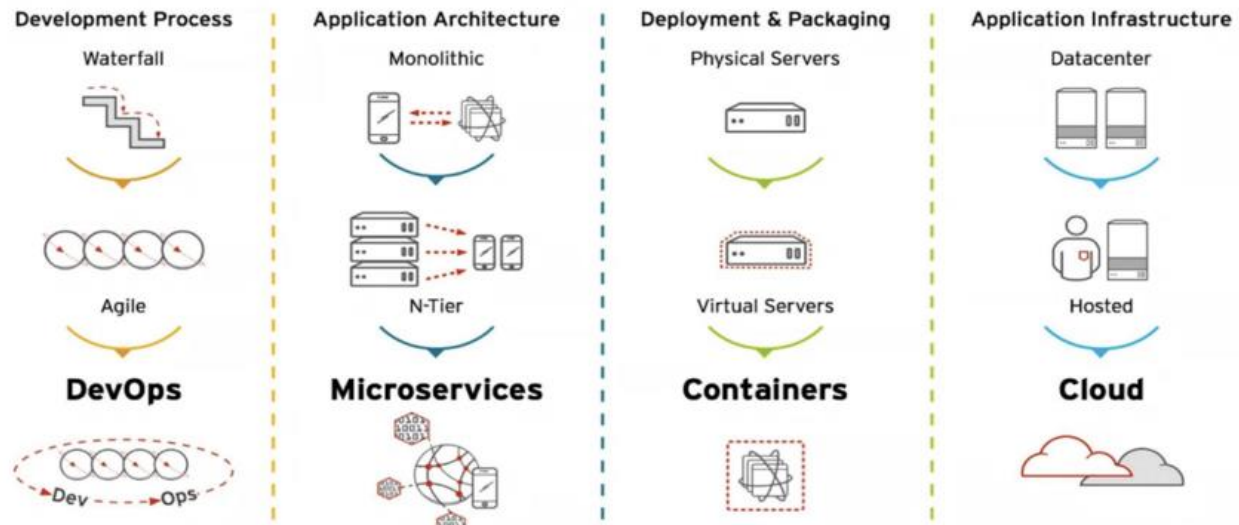
Challenges:

- Infrastructure Readiness
- Scalability
- Quality of product
 - Automation
- Limited Release windows
 - Release may slip/fail
- We want no downtime deployment
 - CICD



Evolution of IT





DevOps:

DevOps is a re-org, not a new team to hire.

Technically....

"DevOps is agile IT operations delivery, required to match the cadence of agile IT development".

"DevOps is the philosophy of unifying Development and Operations at the culture, practice, and tool levels, to achieve accelerated and more frequent deployment of changes to Production (The IT Sceptic)".

DevOps is a software Engineering **Culture** and **Practice** that is unifying Software Development and Software Operations. It Aims at Shorter Deployment cycles, More dependable releases in close alignment with business objectives.

- DevOps is first a Culture of Collaboration between developers and Operations.
- The culture has given scope for set of practices.
- DevOps is a revolutionary movement by Practitioners , for practitioners.

Goals....

- The goal of the development team is to deliver features into the hands of the customer as quickly as possible, so their goal is **speed**. But the goal of the operations team is to maintain the **stability** of the system and minimize downtime.
- stability and speed of delivery
- Time-to-market
- NO Production Failures
- Immediate recovery from failures.
- There is no wall in a DevOps culture.

Container:

Before Container: Pre- configuring our application environment and then deploying our code.

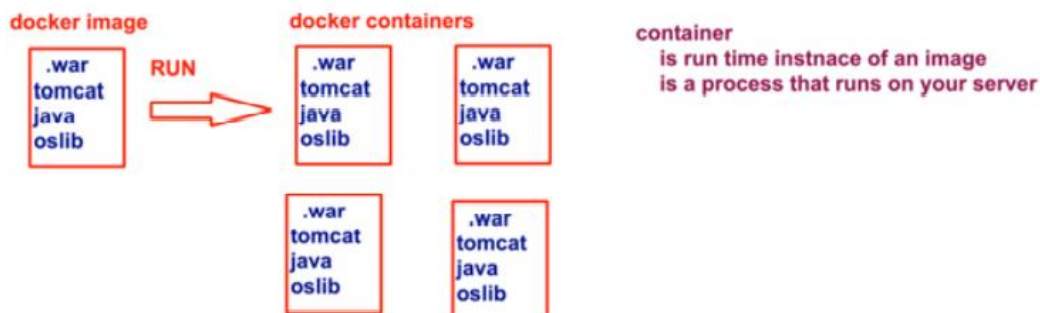
Problem: Application is running on one environment but not running on another environment.

Why? We are just moving artifacts (.war) from one environment to another environment what about dependencies? E.g., OS version, Java version and tomcat version, servers may have inconsistent configuration due to which application is running in one environment but no other environment.

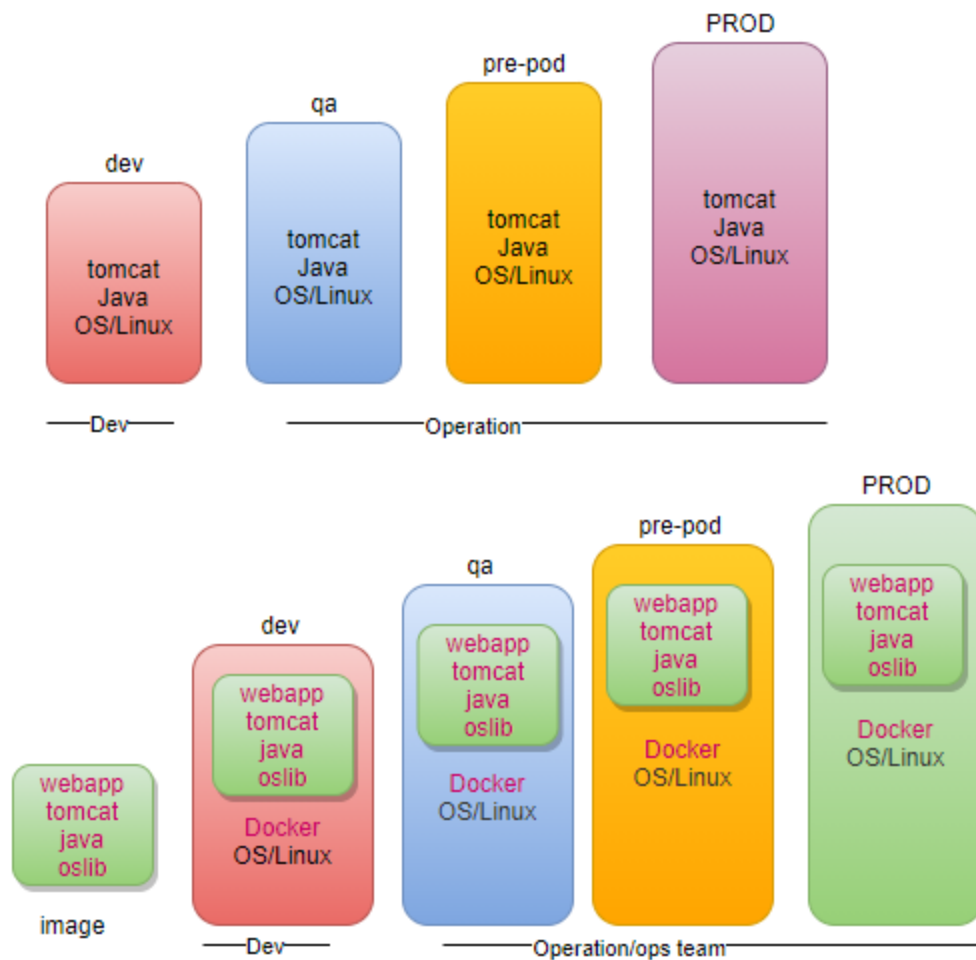
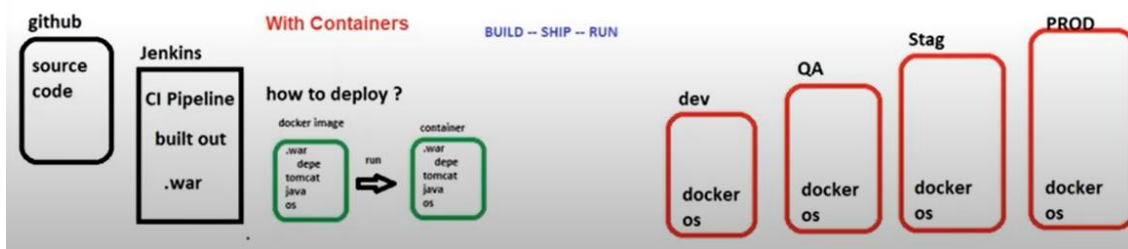
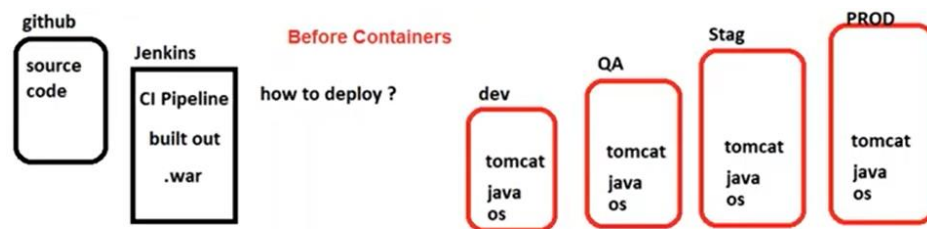
Solution? Container

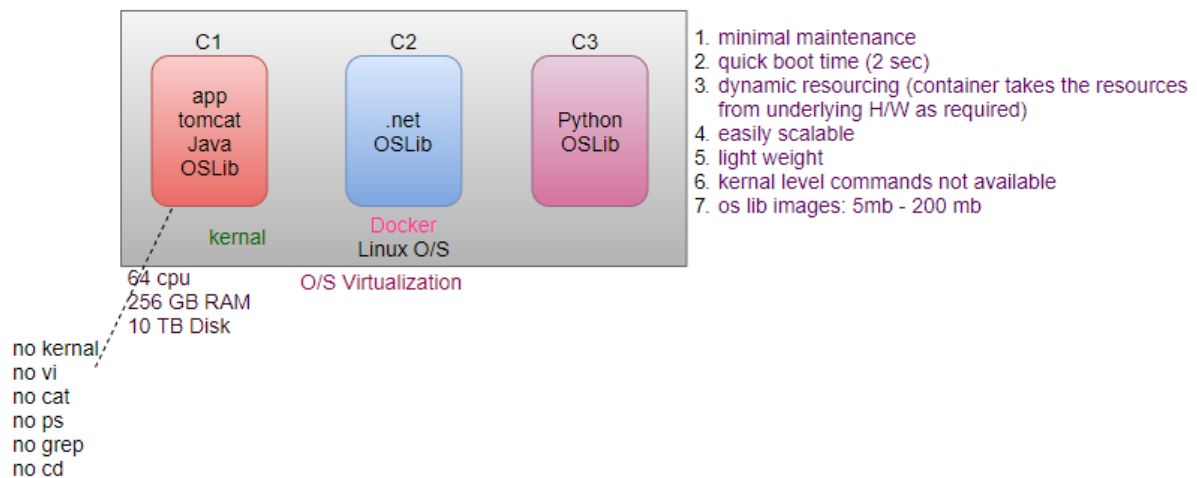
how to build the cont & how to run the cont ?

containerization tools
— helps to build & run containers
ex: docker / containerd / lxc / rocket



Instead of delivering old style, we containerized (bundle the code with all its dependencies into single package and deliver the complete package). This complete package technically called the image.



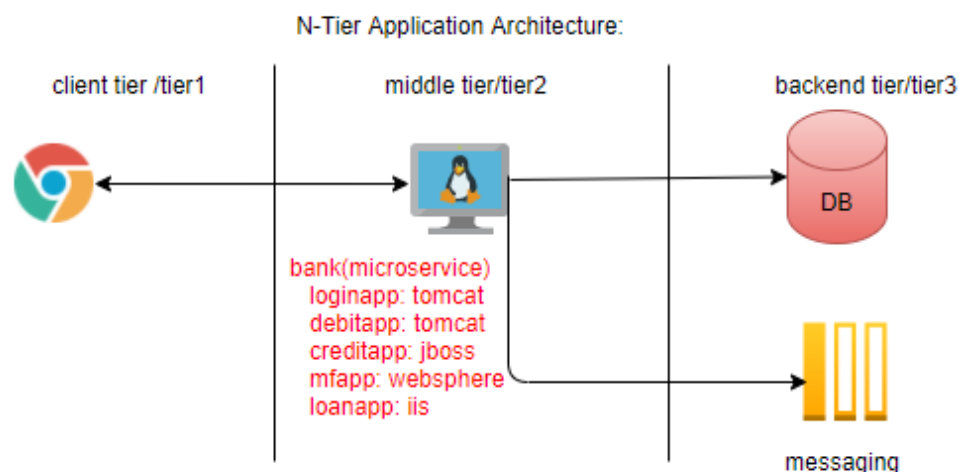


Microservices: (decouple the modules as a separate application)

In our banking application 90% users use debit card and 10% users use credit card, suddenly everything is working fine except the credit card, there is issue in credit card, whole application will be down which will impact the debit card users also be impacted.

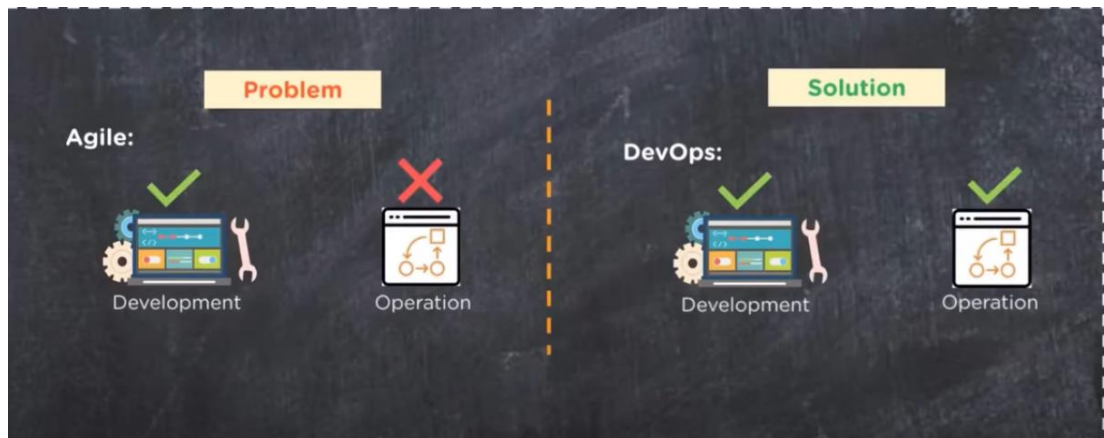
Solution: deploy the modules as a separate application and each application deployed separately on different servers, light weight application can transfer and deploy smoothly, debit card users will not be impacted.

- Technology agnostic.
- Light weight application easy to deliver.
- Easily deployable.
- Debugging is very easy.



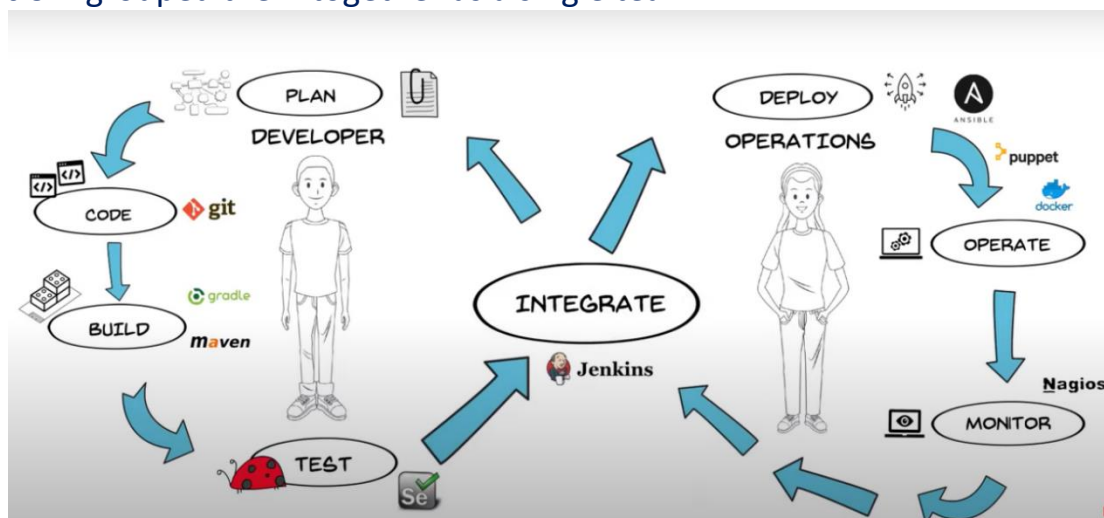
Cloud:

Online delivery, pay as you go. It completely service model. It's like an online food delivery everything available at your home.



Problem Statement: We already have development team, operation team but are not communication properly.

Solution: grouped them together as a single team.



DevOps Stages and Tools

Stage1-1 Plan	JIRA, Rally
Stage-2 Version Control (SCM/VCS)	Git/GitHub, SVN, CVS
Stage-3 Build Code	Maven
Stage-3 Continuous Integration	Jenkins, Travis, TeamCity
Stage-4 Continuous Testing	Jenkins, Selenium, TestNG, Junit
Stage-5 Containerization	Docker
Stage-6 Container Orchestrator	Docker Swarm/ Kubernetes

Stage-7 Configuration Management	Ansible/puppet/Chef
Stage-8 Continuous Monitoring	Splunk, EFK Stack, Nagios

CI/CD achieved through combination of all the tools.

Environment and Tools

Dev	
Source Code Management	Git/GitHub/ CodeCommit
Operations	
Build process	Maven (java)/CodeBuild
Build automation	Jenkins/ Codepipeline
Config Management	Ansible/Config
Infra management tool	Terraform/CloudFormation
Containers	Docker
Container Orchestration	Docker Swarm/Kubernetes
Maintenance	
Monitoring	Nagios, Prometheus/Grafana, CloudWatch
Logging	EFK/CloudWatch

What is Lean?

DevOps Engineer role is a mindset, you should not restrict yourself in a single area, always flexible to shift your area here and there.

Open to learn always, open mind to shift left or right.

Waterfall



Agile



DevOps



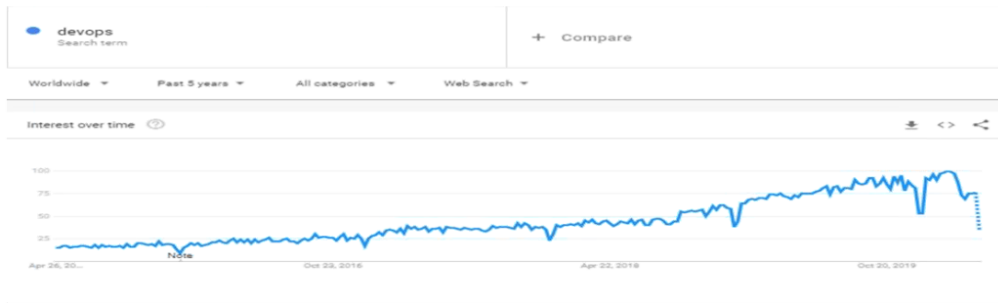
Primary Goal of the DevOps is to create a culture of work together to achieve the speed and stability.

WHY DevOps ?

All Organizations today are working hard to respond better around **changing market needs** and are **keen to adopt DevOps practices**. As a result

All Organizations are now **hiring DevOps Engineers**, who can work & adopt dynamic market needs easily.

a **growth of 75 %** in last couple of years as every organization is adopting DevOps Practices.



DevOps Engineer Roles & Responsibilities ?

With DevOps, there is more scope for frequent changes in the code, due to which it is expected to do

1. Test, build, design, deployment, and ability to maintain **continuous integration and continuous delivery** process using tools like Jenkins, maven, Git, etc.
2. Must know how to choose the best tools and technologies which best fits the business needs.
3. Ability to automate test and deploy the code and monitor.
4. Work in close coordination with the development and operations team such that the application is in line with performance according to the customer's expectation.

It's not expected to Know Everything, but have an idea of end-to-end process, Master couple of tools as Primary Skills then know various tools which can help to achieve the DevOps process.